Disk Granulometries on the Digital Grid: Why a Single Structuring Element Is Not Enough

Peter Zsoldos

Tampere University, Tampere, Finland
peter.zsoldos@tuni.fi

Abstract. Size distributions are usually measured by opening an image with digital disks of increasing radius, as provided by common image processing libraries. This is not a granulometry. As Vogt observed in 1988, digital disks violate Matheron's absorption axiom, and the resulting size distributions can increase with the radius. We find that the violation concerns most pairs of radii and still affects current software, such as DIPlib and the Scientific Python Lectures: on images of disk-shaped particles, their size distributions often increase, so that the pattern spectrum takes negative values. A classical construction repairs this at no extra cost: replacing each opening by the union of the openings by all larger disks gives a correct granulometry that is as round as the disks themselves. The alternative, used by octagons, periodic lines and Bresenham decompositions, keeps a *single* structuring element per size, and these elements are always polygonal. Our main result shows that this cannot be avoided. If the mean radius of a Minkowski chain grows by at most δ per element, every element of radius ρ deviates from every disk by at least about $\rho/(16\pi^2\delta^2)$ in the plane, and by an amount linear in ρ in any dimension. An explicit chain of periodic lines attains this rate up to a factor of 4, and for a fixed library the exact slope is attained by octagons. Experiments on synthetic and real images show when the anisotropy of single-element families matters in practice.

Keywords: Granulometry · Digital disk · Minkowski sum · Lattice polygon · Width function · Pattern spectrum.

1 Introduction

Granulometries, introduced by Matheron [12], formalise sieving. A granulometry is a family of openings $(\gamma_\lambda)_{\lambda\geq 0}$ that decrease with λ and satisfy the absorption law $\gamma_\lambda\gamma_\mu = \gamma_\mu\gamma_\lambda = \gamma_{\max(\lambda,\mu)}$: sieving with two sieves gives the same result as sieving with the coarser one. The size distributions and pattern spectra derived from granulometries [11,17,18] are among the most widely used morphological descriptors. The textbook construction uses one structuring element per size, $\gamma_\lambda(X) = X \circ B_\lambda$. Absorption then holds if and only if B_μ is B_λ-open, i.e. a union of translates of B_λ, whenever $\lambda \leq \mu$. Euclidean disks have this property.

Digital disks do not. Fig. 1(a) shows that the digital disk of radius 5 is not a union of translates of the disk of radius 4, and for the disks $D_r = \{p \in \mathbb{Z}^2 : \|p\| \leq r\}$ the same happens for most pairs of radii. Vogt pointed this out in 1988 [24], and Luengo Hendriks

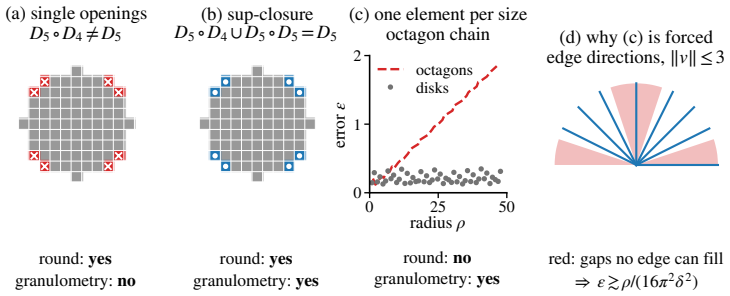


Fig. 1. Overview. (a) Common practice: the red pixels of D_5 lie in no translate of D_4 inside D_5, so absorption fails. (b) The sup-closure also opens by larger disks and recovers them (Proposition 1). (c) One element per size gives a granulometry whose error grows with the radius. (d) This is unavoidable (Theorem 1): short lattice vectors leave gaps of $\arctan(1/\ell)$ next to the axes

et al. [9] accept it as an approximation error. In practice, however, size distributions are still computed as $r \mapsto |X \circ D_r|$. The granulometries of DIPlib [10] and of the Scientific Python Lectures [22] work this way, and on images of round objects their size distributions often increase with the radius, so that the pattern spectrum becomes negative (Sect. 3).

There are two ways to obtain a correct granulometry. The first allows several structuring elements per size. The *sup-closure* $\gamma'_r = \bigvee_{s \geq r} \gamma_{D_s}$, the union of the openings by all disks of radius at least r, is a granulometry for any family (D_s) (Proposition 1). With digital disks it is as round as the disks and costs one opening per radius. The construction is a classical consequence of Matheron's theory [12,17], and opening transforms [23] and local thickness [7] rely on it implicitly, so we do not claim it as new. Our point is a practical one: it is the right way to compute size distributions with disks, and common software does not use it.

The second way keeps a single structuring element per size and builds the elements as *Minkowski chains* $D_k = D_{k-1} \oplus C_k$, which are granulometric by construction [1,8,18,23]. Octagons from neighbourhood sequences [5,15], periodic lines [8], Bresenham segments [1] and Minkowski sums of small digital disks [20] are examples. Each of these constructions improves the roundness of the previous ones, and all of them remain visibly polygonal. The main question of this paper is whether this is a weakness of the known constructions or cannot be avoided. We show that it cannot: a single bound covers all of them, and it is attained. Our contributions are:

- **Diagnosis** (Sect. 3). The violation itself is known [24]. We measure how often it occurs, on which images it changes size distributions, and which current software is affected.
- **Lower bounds** (Theorem 1, Corollaries 1–3). If the mean radius of a Minkowski chain grows by at most δ per element, then *every* element of radius ρ is at Hausdorff distance at least $(1 - o(1))\,\rho/(16\pi^2\delta^2)$ from every Euclidean disk. Bounded steps of the fitted radius force a maximal error of order $\rho^{1/3}$, and for a fixed library the slope $\kappa/(1+\kappa)$ is given by a linear program and attained by octagons.

- **Upper bound and higher dimensions** (Theorems 2 and 3). An explicit chain of periodic lines attains $\Theta(\rho/\delta^2)$ up to a factor of 4, and $\Theta(\sqrt{\rho})$ when $\delta \sim \rho^{1/4}$. In $\mathbb{Z}^n$ the error is at least $c_n \rho\, \delta^{-2n}$.
- **Experiments** (Sect. 6) on synthetic and real images, and recommendations.

Related work. Chamfer norms [2] approximate Euclidean balls by polygons with fixed edge directions, and Corollary 3 applies when their balls form Minkowski chains (e.g. 3×3 masks). Neighbourhood sequences [5,13] and broadcasting sequences [20] optimise the roundness within a fixed library and report errors relative to the radius. Corollary 3 gives the best possible slope for any fixed library, and Theorem 1 shows that letting the library grow with bounded steps cannot beat ρ/δ^2. Luengo Hendriks et al. [9] improve the accuracy of disk sieves with shifted and interpolated disks, for which absorption holds only approximately; the sup-closure applies to such elements as well. Soille and Talbot [19] noted that radial decompositions by Bresenham lines are not granulometric. Normand [14] builds convex structuring elements by unions of translates, which lies outside the scope of our bound. To our knowledge, no lower bound of this kind has been published.

2 Preliminaries

For a finite set $X \subset \mathbb{Z}^n$ and a unit vector u, let $h_X(u) = \max_{p \in X} \langle p, u \rangle$ be the support function and $w_X(u) = h_X(u) + h_X(-u)$ the *width function*, i.e. the width of $\operatorname{conv} X$ in direction u. Both are additive under Minkowski addition [16]:

$$h_{X \oplus Y} = h_X + h_Y, \qquad w_{X \oplus Y} = w_X + w_Y. \tag{1}$$

The *anisotropy* of X is $\mathrm{A}(X) = \max_u w_X - \min_u w_X$, and its *mean radius* is $\bar{r}(X) = \frac{1}{2} \int w_X \, d\sigma$, where σ is the uniform probability measure on the sphere; $\bar{r}$ is additive as well. In the plane, Cauchy's formula gives a concrete meaning: $\bar{r}(X)$ is the perimeter of $\operatorname{conv} X$ divided by 2π, i.e. the radius of the circle with the same perimeter. For a segment of length L it equals L/π. We say that X is *ε-round with radius ρ* if $d_H(\operatorname{conv} X, B(c, \rho)) \le \varepsilon$ for some centre c. Then $|h_X(u) - \langle c, u \rangle - \rho| \le \varepsilon$, and hence

$$2\rho - 2\varepsilon \le w_X \le 2\rho + 2\varepsilon, \qquad \mathrm{A}(X) \le 4\varepsilon, \qquad |\bar{r}(X) - \rho| \le \varepsilon. \tag{2}$$

The opening of Y by a structuring element D is $Y \circ D = \bigcup \{D + t : D + t \subseteq Y\}$. A family (D_k) generates a *single-element granulometry* $\gamma_k(Y) = Y \circ D_k$ if and only if $D_l \circ D_k = D_l$ for all $k \le l$ [12,17]; since the relation is transitive, consecutive elements suffice. A *Minkowski chain* is a family with $D_k = D_{k-1} \oplus C_k$, and every Minkowski chain generates a single-element granulometry. Its *increments* are the C_k, its *steps* are $\bar{r}(D_k) - \bar{r}(D_{k-1}) = \bar{r}(C_k)$, and, if all C_k belong to a finite set $\mathcal{L}$ (up to translation), $\mathcal{L}$ is its *library*.

The following fact is classical [12,17]; we include a short proof for completeness.

Proposition 1 (sup-closure). *For any finite sets $D_1, \dots, D_R \subset \mathbb{Z}^n$, the operators $\gamma'_r = \bigvee_{s=r}^{R} \gamma_{D_s}$, $r = 1, \dots, R$, where $\gamma_D(Y) = Y \circ D$, form a granulometry.*

Proof. A supremum of openings is an opening [17], and γ'_r is decreasing in r. Let $r \le s$. As both are anti-extensive and increasing, $\gamma'_r\gamma'_s$ and $\gamma'_s\gamma'_r$ are $\le \gamma'_s$; both are $\ge \gamma_{D_t}\gamma_{D_t} = \gamma_{D_t}$ for every $t \ge s$, hence $\ge \gamma'_s$. □

Computing γ'_r from $r = R$ down to 1 needs one opening per radius, the same cost as the single-element family γ_{D_r}; for Gauss disks, each opening is computed exactly by two Euclidean distance transforms, $Y \ominus D_r = \{\mathrm{EDT}_Y > r\}$ and $Z \oplus D_r = \{\mathrm{EDT}_{Z^c} \le r\}$, at a cost independent of r. Since flat openings commute with thresholding, the violation and its repair carry over to grey-level images. The structuring elements are unchanged, so with digital disks the sup-closure is exactly as round as the disks. Correctness and roundness can therefore be had together, as long as several elements per size are allowed. The next section shows how often single disk openings fail; Sections 4 and 5 then show what is lost when one insists on a single element per size.

3 Digital Disks Are Not Granulometric

The example of Fig. 1(a) is not an exception. We tested the condition "D_r is D_s-open" for all pairs $1 \le s < r \le 40$, for the digital disks of `scikit-image` [25] (`disk`, the Gauss digitisation $\{p : \|p\| \le r\}$) and of OpenCV [3] (`MORPH_ELLIPSE`). As Table 1 shows, most pairs violate it, including most consecutive pairs. When D_r is not D_s-open, $X \circ D_r \not\subseteq X \circ D_s$ for some X, and the size distribution $r \mapsto |X \circ D_r|$ can *increase*, so the pattern spectrum takes negative values. On random 120×120 images of 3 to 11 overlapping disks or rectangles ($r \le 15$), this happens for 63% (scikit-image) and 60% (OpenCV) of the images; Fig. 3(a,b) shows a pronounced example. `scipy.ndimage`, `scikit-image` and OpenCV give the same results up to border conventions.

Current software. The source code of widely used tools (October 2026) shows the common practice and both correct routes. The isotropic `Granulometry` of DIPlib 3.6.1 [10], the granulometry example of the Scientific Python Lectures [22] and the MATLAB example [21] apply one opening per scale by a sampled disk. The morphology library Smil [4] opens by homothetic hexagons or squares, i.e. a Minkowski chain, which is correct but polygonal. PoreSpy [6] derives size distributions from the local thickness, which labels each pixel with the largest disk that covers it; this is exactly the sup-closure. On the 400 random images above, the size distributions of the Lectures code increase on 253 images, as with `disk`, and those of DIPlib on 155 images for diameters $3, 5, \ldots, 31$ and on 48 images with its default settings (12 scales spaced by $\sqrt{2}$). We could not test MATLAB.

When does it matter? The increases come from round objects; rectangles never produced one. They are most frequent when the objects are digital disks themselves, as in rendered or simulated scenes: with integer centres and radii, 92% of the images are affected, and the negative part of the pattern spectrum has a median of 2.6% (at most 14%) of its positive part. For disks at real-valued positions, as for real particles, 31% of the images are affected, but only by 0.3% (median). On the natural sample images of scikit-image and on random 3D packings of balls, we found almost no increases. The violation is thus rarely large. It does, however, break the guarantee that a size distribution is monotone and

Table 1. Standard digital disks violate the absorption condition ($1 \le s < r \le 40$), and size distributions computed with them increase on most random images. The sup-closure (Proposition 1) never does

	pairs with D_r not D_s-open		images with an
family	all $s < r$	$s = r - 1$	increasing $\lvert X \circ D_r \rvert$
`scikit-image disk`	571/780 (73%)	30/39	253/400
OpenCV `MORPH_ELLIPSE`	628/780 (81%)	36/39	241/400
sup-closure of `disk`	–	–	0/400

a pattern spectrum non-negative, on which normalising the spectrum into a probability distribution, or computing its moments and entropy, relies, and the repair costs nothing.

4 Lower Bounds for Single-Element Chains

Why not simply choose better elements for a single-element family? For Minkowski chains, the answer lies in additivity (1): it turns the roundness of two elements of a chain into a constraint on the increments between them, and all edges of the increments are lattice vectors. If the increments are short, these vectors leave a gap of directions around each axis (Fig. 1d), and a polygon whose edges avoid an angular interval cannot be round: over the gap, a single vertex carries the support and sticks out of the circle. Adding more elements to the chain makes the polygon larger but does not fill the gap, so the deviation grows with the radius.

Lemma 1 (direction gap). *Let $\ell \ge 1$. No nonzero $(a, b) \in \mathbb{Z}^2$ with $a^2 + b^2 \le \ell^2$ has a direction angle in $(0, \arctan(1/\ell))$.*

Proof. A direction in that interval requires $a > 0$ and $b \ge 1$, hence $b/a \ge 1/a \ge 1/\ell$. □

Theorem 1. *Let $(D_k)_{k=0}^{K}$ be a Minkowski chain in $\mathbb{Z}^2$ whose increments have diameter at most $\ell \ge 1$. If D_0 is ε_0-round with radius ρ_0 and D_K is ε_K-round with radius ρ_K, then, with $c = 1 - \cos\big(\frac{1}{2}\arctan\frac{1}{\ell}\big)$,*

$$4(\varepsilon_0 + \varepsilon_K) \;\ge\; 2\,(\rho_K - \rho_0 - \varepsilon_0 - \varepsilon_K)\,c. \tag{3}$$

Proof. Step 1. Every edge of $\operatorname{conv} C_k$ joins two lattice points, so it is a lattice vector of norm at most ℓ. *Step 2.* Let $P = \sum_k \operatorname{conv} C_k$ and $Q = \frac{1}{2}(P - P)$. By (1), $W := w_{D_K} - w_{D_0} = w_P = 2h_Q$. The edges of the centrally symmetric polygon Q are parallel to edges of the $\operatorname{conv} C_k$, so by Lemma 1 (applied to $\pm e$) none has a direction in $(0, \alpha)$, where $\alpha = \arctan(1/\ell)$. The normal angles in $(\frac{\pi}{2}, \frac{\pi}{2} + \alpha)$ therefore belong to the normal cone of a single vertex q of Q: $h_Q(\theta) = |q|\cos(\theta - \theta_q)$ on an interval I of length at least α. *Step 3.* We may assume $s := \rho_K - \rho_0 - \varepsilon_0 - \varepsilon_K > 0$, since otherwise (3) is trivial. By (2), $\min W \ge 2s$, so $h_Q > 0$ and $|\theta - \theta_q| < \frac{\pi}{2}$ on I. On an interval of length α within $(\theta_q - \frac{\pi}{2}, \theta_q + \frac{\pi}{2})$, the function $\cos(\theta - \theta_q)$ varies by at least

$1-\cos(\alpha/2)$: by at least that much if $\theta_q \in I$, and by at least $1-\cos\alpha$ otherwise, by monotonicity. With $|q| \geq \min h_Q \geq s$ this gives $\max W - \min W \geq 2sc$. *Step 4.* Finally $\max W - \min W \leq \mathrm{A}(D_K) + \mathrm{A}(D_0) \leq 4\varepsilon_K + 4\varepsilon_0$ by (2). □

For a chain starting from a single point ($\varepsilon_0 = \rho_0 = 0$), (3) gives $\varepsilon_K \geq \rho_K\, c/(2+c)$, where $c = \frac{1}{8\ell^2}(1 + O(\ell^{-2}))$: *every* element is at least $(1-o(1))\,\rho/(16\ell^2)$ away from a disk. Lemma 1 uses only the lattice: in any planar lattice Λ with a shortest vector v, no vector $w \in \Lambda$ with $\|w\| \leq \ell$ makes an angle in $(0, \arcsin(\det\Lambda/(\|v\|\ell)))$ with v, since $|\det(v,w)|$ is a nonzero multiple of $\det\Lambda$. Theorem 1 therefore holds on any planar grid, e.g. the hexagonal one, with this angle in place of $\arctan(1/\ell)$. Different constraints on the steps bound ℓ in different ways.

Corollary 1 (mean-radius steps). *If a Minkowski chain starting from a point has mean-radius steps $\bar r(C_k) \leq \delta$ (its size resolution), then $\ell \leq \pi\delta$, and every element of radius ρ has error $\varepsilon \geq \rho\, c/(2+c) = (1-o(1))\,\rho/(16\pi^2\delta^2)$, with $c = 1-\cos(\frac{1}{2}\arctan\frac{1}{\pi\delta})$. In particular, granulometric chains with bounded steps have error growing linearly with the radius.*

Proof. By Cauchy's formula the perimeter of a planar convex set is π times its mean width, and it is at least twice its diameter. Hence $\operatorname{diam}\operatorname{conv} C_k \leq \frac{\pi}{2}\cdot 2\bar r(C_k) \leq \pi\delta$. □

Corollary 2 (fitted-radius steps). *If all elements of a Minkowski chain are ε-round with radii satisfying $\rho_k - \rho_{k-1} \leq \Delta$, then $\varepsilon \geq (1-o(1))\,((\rho_K-\rho_0)/512)^{1/3}$ as $\rho_K \to \infty$.*

Proof. By (1) and (2), $w_{C_k} = w_{D_k} - w_{D_{k-1}} \leq 2\Delta + 4\varepsilon$, so $\ell \leq 2\Delta + 4\varepsilon$. If ε stayed bounded, then so would ℓ, and (3) would force $\varepsilon \to \infty$. Hence $\ell \leq 4\varepsilon(1+o(1))$, and (3) with $\varepsilon_0 = \varepsilon_K = \varepsilon$ becomes $8\varepsilon \geq (\rho_K-\rho_0)/(64\varepsilon^2)\,(1+o(1))$. □

Corollary 3 (fixed library). *Let $\mathcal{L}$ be a finite library containing a set with more than one point, and let*

$$\kappa(\mathcal{L}) = \inf\Big\{\tfrac{1}{4}\,\mathrm{A}\big(\textstyle\sum_C \lambda_C w_C\big) : \lambda \geq 0,\ \min_u \sum_C \lambda_C w_C(u) = 2\Big\}.$$

Then $\kappa := \kappa(\mathcal{L}) > 0$, and the last element of every chain with library $\mathcal{L}$ satisfies $\varepsilon_K \geq \big(\kappa(\rho_K - \rho_0 - \varepsilon_0) - \frac{1}{4}\,\mathrm{A}(D_0)\big)/(1+\kappa)$, i.e. $\varepsilon \geq \frac{\kappa}{1+\kappa}\rho - O(1)$.

Proof. For λ in the simplex, $f(\lambda) = \mathrm{A}(\sum\lambda_C w_C)/\min\sum\lambda_C w_C$ takes values in $(0,\infty]$: $\sum\lambda_C w_C$ is the width function of the polygon $\sum\lambda_C \operatorname{conv} C$, which does not have constant width, because bodies of constant width are strictly convex [16]. Moreover f is continuous where finite and tends to ∞ where the minimum vanishes, so it attains a positive minimum, and $\kappa = \frac{1}{2}\min f > 0$. For the chain, write $W = w_{D_K} - w_{D_0} = \sum_C n_C w_C$ and $s = \frac{1}{2}\min W \geq \rho_K - \rho_0 - \varepsilon_0 - \varepsilon_K$. Then $\mathrm{A}(W) \geq 4\kappa s$ and $\mathrm{A}(W) \leq 4\varepsilon_K + \mathrm{A}(D_0)$. □

We compute κ by linear programming on a grid of 3600 directions, with all D_4-images included; the grid underestimates the anisotropy, so the computed values are lower approximations and the bounds remain valid. It is 0.2071 for squares ($\{N_4\}$ or $\{N_8\}$; exactly $(\sqrt{2}-1)/2$) and 0.0412 for octagons, and it decreases for richer libraries, e.g. to 0.0028 for the 8-connected digital straight segments (DSS) of all primitive v with $\|v\| \leq 10$. Corollary 3 is tight for octagons: the element of fitted radius 961.5 of the best octagon chain has $\varepsilon/\rho = 0.039571$, while $\kappa/(1+\kappa) = 0.039570$.

5 Matching Upper Bound and Higher Dimensions

How round can a single-element chain be? The lower bound is attained, up to constants, by a simple explicit chain. For $m \geq 1$, let V_m contain one vector from each pair $\pm v$ of integer points on the boundary of $[-m, m]^2$. These are $4m$ vectors of norm at most $\sqrt{2}m$, and consecutive directions differ by at most $\arctan(1/m) \leq 1/m$, the largest gap being between $(m, 0)$ and $(m, 1)$. Let Z be the polygon circumscribed to the unit disk whose edges have the directions of $\pm V_m$; it is centrally symmetric, hence $Z = \sum_{v \in V_m} [-\frac{L_v}{2}\hat{v}, \frac{L_v}{2}\hat{v}]$ with $\hat{v} = v/\|v\|$, and $1 \leq h_Z \leq \sec(\frac{1}{2m})$. For $t \geq 0$ set $n_v(t) = \lfloor tL_v/\|v\| \rfloor$ and

$$D(t) = \bigoplus_{v \in V_m} n_v(t)\{0, v\},$$

the Minkowski sum of $n_v(t)$ copies of the periodic line $\{0, v\}$ [8]. As t grows, the counts increase. When several counts increase at the same t, we add the corresponding lines one at a time, so the elements form a Minkowski chain of periodic lines starting from $\{0\}$.

Theorem 2. *This chain has mean-radius steps at most $\sqrt{2}m/\pi$, and every element is ε-round with some radius ρ, with $\varepsilon \leq \rho/(8m^2) + 2m^2$.*

Proof. A step adds one periodic line, whose mean radius is $\|v\|/\pi \leq \sqrt{2}m/\pi$. Every element, including those created when ties are broken, satisfies $n_v = tL_v/\|v\| - \varphi_v$ for some t and $\varphi_v \in [0, 1]$. The centred support function of $\operatorname{conv} D = \sum_v n_v[0, v]$ is therefore $H = \sum_v n_v|\langle v, u\rangle|/2 = t\, h_Z - \sum_v \varphi_v|\langle v, u\rangle|/2$. Since $\sum_v \|v\| \leq 4\sqrt{2}m^2$ and $\sec y - 1 \leq y^2$ for $y \leq \frac{1}{2}$, we get $t - 2\sqrt{2}m^2 \leq H \leq t + t/(4m^2)$. The disk of radius $\rho = (\max H + \min H)/2 \geq t - 2\sqrt{2}m^2$ is thus within $\varepsilon \leq t/(8m^2) + \sqrt{2}m^2 \leq \rho/(8m^2) + \sqrt{2}m^2 + \frac{\sqrt{2}}{4} \leq \rho/(8m^2) + 2m^2$. □

With $\delta = \sqrt{2}m/\pi$, Theorem 2 gives $\varepsilon \leq \rho/(4\pi^2\delta^2) + \pi^2\delta^2$, while Corollary 1 gives $\varepsilon \geq (1 - o(1))\, \rho/(16\pi^2\delta^2)$. For $\rho \geq 4\pi^4\delta^4$, where the constant term is dominated, the best achievable error is therefore $\Theta(\rho/\delta^2)$, with constants within a factor of 8, and of 4 asymptotically. We can also concatenate chains with $m_j = 2^j$, each run up to $t_j \propto 16^j$ on top of the frozen previous ones; anisotropies and radii add. The result is a single chain with steps $O(\rho^{1/4})$ and error $O(\sqrt{\rho})$, which is optimal for steps $O(\rho^{1/4})$ by Corollary 1. Numerically, the chain with $m = 2$ settles at $\varepsilon \approx \rho/(18m^2)$, between the asymptotic bounds $\rho/(32m^2)$ and $\rho/(8m^2)$ (Fig. 2b); larger m converge more slowly. The multiscale chain satisfies $\varepsilon \leq 0.28\sqrt{\rho}$ for $20 \leq \rho \leq 5 \cdot 10^4$ (Fig. 2c).

Theorem 2 concerns convex hulls. Sums of periodic lines can leave isolated holes [8]; adding the 2×2 square reduces them to 0.2–2% of the hull's lattice points, and they do not affect the granulometric property.

Higher dimensions. In $\mathbb{Z}^n$ the directions are not ordered, but a measure argument replaces Lemma 1. The constants are explicit, but the exponent is not sharp: in the plane, the same argument gives ℓ^{-4} instead of ℓ^{-2}.

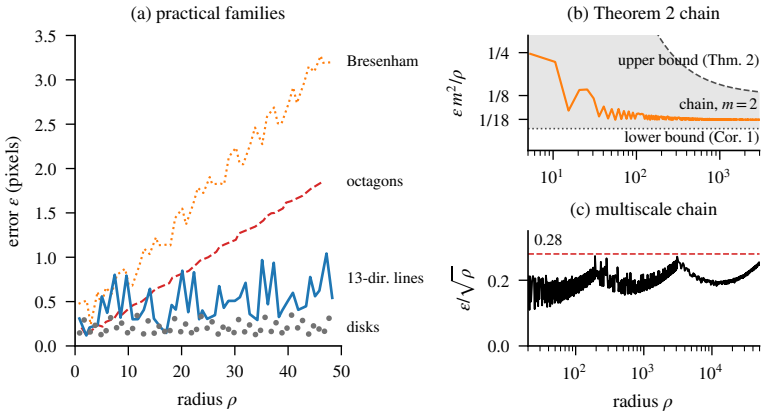


Fig. 2. Roundness error ε of single-element chains. (a) Practical families (all in Table 2); the disks used by the sup-closure for reference. (b) The chain of Theorem 2 with $m = 2$, normalised by ρ/m^2, settles at about $1/18$ between the bounds of Theorem 2 and Corollary 1. (c) The multiscale chain, normalised by $\sqrt{\rho}$, stays below 0.28

Theorem 3. *Let (D_k) be a Minkowski chain in $\mathbb{Z}^n$, $n \geq 3$, whose increments have diameter at most $\ell \geq 1$, and let D_0 and D_K be ε_0- and ε_K-round with radii ρ_0 and ρ_K (with respect to Euclidean balls). Let $b_n = 2\Gamma(\frac{n}{2})/(\sqrt{\pi}\,\Gamma(\frac{n-1}{2}))$. Then (3) holds with c replaced by $1 - \cos\alpha_n$, where $\sin\alpha_n = 1/(4b_n(2\ell+1)^n)$. Hence $\varepsilon_K \geq (\rho_K - \rho_0 - \varepsilon_0)/(96 \cdot 9^n b_n^2 \ell^{2n}) - \varepsilon_0$. If the mean-radius steps are at most δ, then $\ell \leq 2\delta/\mu_n$ with $\mu_n = \Gamma(\frac{n}{2})/(\sqrt{\pi}\,\Gamma(\frac{n+1}{2}))$, and bounded steps again force linear growth.*

Proof. The directions u split into regions in each of which a single vertex of $Q = \frac{1}{2}(P-P)$ attains $h_Q(u)$ (the normal fan of Q). The boundaries between these regions lie in the hyperplanes $e^\perp$ for the edges e of the conv C_k, i.e. lattice vectors with $\|e\| \leq \ell$. There are at most $(2\ell+1)^n$ of them, and each band $\{u : |\langle u, e/\|e\|\rangle| < s\}$ of the sphere has measure at most $b_n s$, since for $n \geq 3$ the density of $\langle u, e/\|e\|\rangle$ is maximal at 0. With $s = 2\sin\alpha_n$ the bands cover at most half of the sphere, so some u_0 satisfies $|\langle u_0, e\rangle| \geq 2\sin\alpha_n \|e\|$ for all e, and the geodesic cap of radius α_n around u_0 meets no boundary. On this cap $h_Q(u) = \langle q, u\rangle$ is linear. Restricting it to the great circle through u_0 and q gives a variation of at least $|q|(1 - \cos\alpha_n)$, and Steps 3 and 4 are as before. The explicit bound follows from $1 - \cos\alpha_n \geq \frac{1}{2}\sin^2\alpha_n$ and $2\ell + 1 \leq 3\ell$. Finally, a segment of length L has mean width $\mu_n L$, and mean width is monotone under inclusion. □

6 Experiments

The bounds say how far single-element families must deviate from disks; the experiments ask whether this matters for real measurements.

Families. We compare Gauss disks (`disk`), their sup-closure (Proposition 1), and five single-element families with largest mean-radius steps between 1.4 and 2.2, i.e. about

Table 2. Roundness for $\rho \leq 48$ (error ε in pixels); last column: largest radius with $\varepsilon \leq 1$. The sup-closure uses several elements per size, the others one

	granulo-	largest	error ε		radius with
family	metry	step δ	max	mean	$\varepsilon \leq 1$
Gauss disks (single)	no	1.19	0.35	0.21	–
sup-closure of Gauss disks	yes	1.19	0.35	0.21	≥ 48
octagons	yes	2.17	1.85	0.98	25
Bresenham/DSS, 8 dir.	yes	1.48	3.27	1.75	12
periodic lines, 8 dir.	yes	1.42	1.09	0.60	41
periodic lines, 13 dir.	yes	1.46	1.04	0.48	46
DSS chain, MILP	yes	2.09	1.00	0.57	44

one element per integer radius. These are octagons; Bresenham/DSS segments as in [1]; periodic lines as in [8] in 8 and in 13 directions (the D_4-images of $(1,0)$, $(1,1)$, $(2,1)$ and, for 13, also $(3,1)$); and a chain of individual DSS computed by mixed-integer programming (HiGHS solver), which minimises the maximal error for $r \leq 48$. Segment libraries use the optimal weights λ^* of Corollary 3, rounded greedily to multiplicities $\approx t\lambda^*$. These chains contain the 2×2 square, and all their elements are hole-free.

Table 2 and Fig. 2(a) summarise roundness. Among single-element families, the optimised chains halve the error of octagons and reach a third of that of Bresenham decompositions, but they remain three times less round than the disks used by the sup-closure. At these radii Corollary 1 only forces 0.06–0.14: the force of the bound lies in the growth rate (Fig. 2b). As Theorem 1 predicts through ℓ, larger steps help: with radius steps up to 4.1, the 13-direction chain reaches 0.41.

Size distributions under rotation. For each scene X, rotated by every multiple of 5° up to 45°, we compute $F_\theta(r) = |\gamma_r(X_\theta)|/|X_\theta|$. We report the rotation sensitivity $S = \mathrm{mean}_r\, \mathrm{std}_\theta\, F_\theta(r)$ and the number of increases $F_\theta(r+1) > F_\theta(r)$, which are impossible for a granulometry. The scenes are a synthetic image of nine squares of sides 17 to 81 pixels, rasterised exactly after rotation ($r \leq 40$), and three real images from `scikit-image`: nuclei (`human_mitosis`, $r \leq 15$), `coins` ($r \leq 30$), and a material texture (`gravel`, $r \leq 20$). The real images are smoothed, rotated with bilinear interpolation, and thresholded at the Otsu level of the unrotated image.

Per-object sieve size. Sieving assigns to each object O the size $s(O) = \max\{r : \gamma_r(X) \cap O \neq \emptyset\}$ of the largest element that fits inside it, which is the same for Gauss disks and their sup-closure. We compare $s(O)$ with the Euclidean inscribed radius, the maximum of the distance transform in O. We report the mean error and rotation standard deviation of $s(O)$ relative to the inscribed radius, excluding one merged object larger than the tested radii.

Results. On every scene, the sup-closure has exactly the rotation sensitivity of the Gauss disks but never increases (Table 3, Fig. 3b,c): it removes the 17 increases of

Table 3. Behaviour under rotation; lower is better everywhere. Left: rotation sensitivity S ($\times 10^3$) of the size distributions, with the number of increases in parentheses when nonzero. Right: sieve size of the 21 objects of `coins` (inscribed radius 18–30 pixels), in % of the inscribed radius

	rotation sensitivity S ($\times 10^3$)				`coins`, per object	
family	squares	nuclei	coins	gravel	error	rot. std
Gauss disks (single)	4.0 (17)	1.7	5.0	1.4	2.8	0.65
sup-closure of Gauss disks	4.0	1.7	5.0	1.4	2.8	0.65
octagons	30.2	1.9	5.0	1.3	4.8	1.13
Bresenham/DSS, 8 dir.	13.3	1.9	6.4	1.6	5.7	1.20
periodic lines, 8 dir.	11.8	2.0	4.2	1.1	4.6	1.06
periodic lines, 13 dir.	9.5	1.6	5.0	1.6	4.1	0.54
DSS chain, MILP	17.6	1.4	6.2	1.2	3.8	1.00

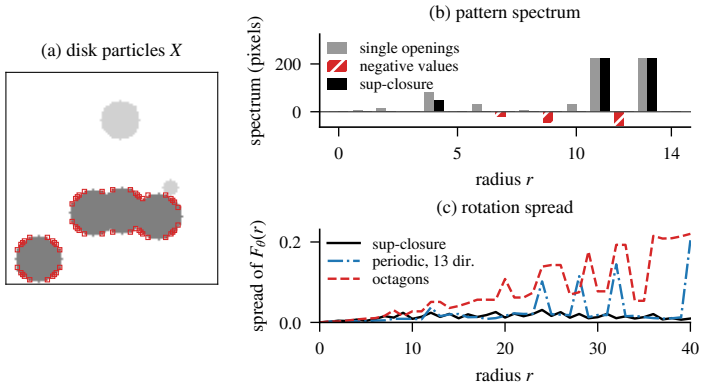


Fig. 3. (a) Disk particles X (light grey), $X \circ D_{12}$ (dark grey), and pixels of $X \circ D_{13}$ not in $X \circ D_{12}$ (red). (b) Pattern spectrum of X: single disk openings give negative values, the sup-closure does not (two bars clipped). (c) Rotation spread $\max_\theta F_\theta - \min_\theta F_\theta$ on the squares scene

single openings on the squares scene, and all increases on the random images of Sect. 3 (0 of 400 images, against 253).

Single-element families pay for their polygonal shape. On the squares scene (Fig. 3c), octagons are 7.5 times more orientation dependent than the disks, and the 13-direction periodic chain reduces this to 2.4 times. Per object on `coins`, octagons have almost twice the size error and rotation scatter of the disks, and the 13-direction chain has the smallest scatter (Table 3). On small objects, the shape of the element hardly matters. On nuclei and gravel, with inscribed radii of 3–10 pixels, discretisation dominates: every family errs by 13–24%, and none is consistently best.

Recommendations. To compute size distributions or pattern spectra with disks, one should not open by each library disk separately. The sup-closure of Proposition 1 opens by every radius from the largest down and accumulates the unions. It costs nothing extra and yields a correct granulometry that is as round as the disks. A single

structuring element per size is needed only where distance transforms are not available and each opening must be a fixed cascade of two-point erosions and dilations, as in some hardware and streaming implementations. In that case, a Minkowski chain with LP weights improves on octagons and Bresenham decompositions, but not on the sup-closure: up to radius 48 the 13-direction periodic chain has error at most 1.04 pixels, and larger radii need larger libraries (Fig. 2b). The code and the computed chains are available at https://github.com/Peter-Zsoldos/dgmm2027-granulometry.

7 Conclusion

Opening by each digital disk of a library is not a granulometry. This has been known since 1988, yet it still makes the size distributions of current software increase. The classical sup-closure fixes the problem at no cost and should be used instead. Families with a single element per size, which decades of constructions have tried to make rounder, cannot do as well: with mean-radius steps δ, their error in the plane is $\Theta(\rho/\delta^2)$, and bounded steps force linear growth in every dimension. Open questions are the optimal rate when only the fitted radius has bounded steps (between $\rho^{1/3}$ and $\rho^{1/2}$), whether single-element granulometries that are not Minkowski chains [14] can escape the bound, and matching upper bounds in $\mathbb{Z}^3$.

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